

Coffee Shop Management System

Software Requirement Specification

Version 1.0

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1. Introduction:

• 1.1 Purpose:

This project focuses on designing a coffee shop database system to organize and manage data such as employees, coffee beans, orders, customers, equipment and roasters. It helps improve data organization and supports better management of coffee shop operations.

• 1.2 Scope:

• Data Storage (Store):

Saving structured data in a relational database system.

• Data Organization (Manage):

Organizing data into related tables such as employees, customers, equipment, skills, beans, and orders.

• Relationship Management:

Defining relationships between entities (e.g., employee skills, orders, and items).

• Data Retrieval:

Allowing efficient access to stored data using queries.

• Reporting:

Providing organized data that can be used for analysis or dashboards.

• 1.3 Overview:

This project focuses on building a database system for a coffee shop in a clear and simple way. The system helps organize data and supports better understanding of operations.

2. General Description

• 2.1 Users:

•The system can be useful for:

- ✓ Coffee shop managers
- ✓ Employees (Baristas)
- ✓ System administrators
- ✓ Non-technical users who want organized data

• 2.2 Objectives:

- The project aims to:
 - ❖ Organize coffee shop data efficiently
 - ❖ Manage employee information and skills
 - ❖ Track coffee beans and their details
 - ❖ Manage customer orders and items
 - ❖ Support better decision-making

3. Data Requirements

➤ Database tables include:

- Employee
- Skill
- Employee_Skill
- Bean
- Bean_FlavorNotes
- Roaster
- Orders
- Order_Item
- Customer
- Equipments

➤ Data includes:

- Employee information (Name, Email, Phone)
- Skills and relationships
- Coffee beans and flavor notes
- Roaster location (City, Country)
- Customer information (ID, Name, phone)
- Equipments (ID)
- Orders and order details

➤ Data should be structured and consistent

4. Functional Requirements

• 4.1 the system should:

- Add employee data
- Update employee information
- Link employees with skills
- Add coffee beans and roasters
- Store flavor notes for each bean

- Create new orders
- Add items to each order
- Store quantity and price
- Retrieve data using queries
- Maintain relationships between tables

5. Non-Functional Requirements

- Data should be accurate
- System should be reliable
- Queries should run efficiently
- Database design should be organized
- System should be easy to understand

6. Tools & Technologies

• **6.1 Database System:** MYSQL

• **6.2 Design Tools:** Draw.io

• **6.3 Platform:** SQL environment

7. Expected Output

• **7.1 the project is expected to produce:**

- A well-structured database
- ER Diagram (Conceptual Design)
- Logical schema (tables and relationships)
- Organized data table
- A simple prototype (database structure visualization)
- Clear documentation

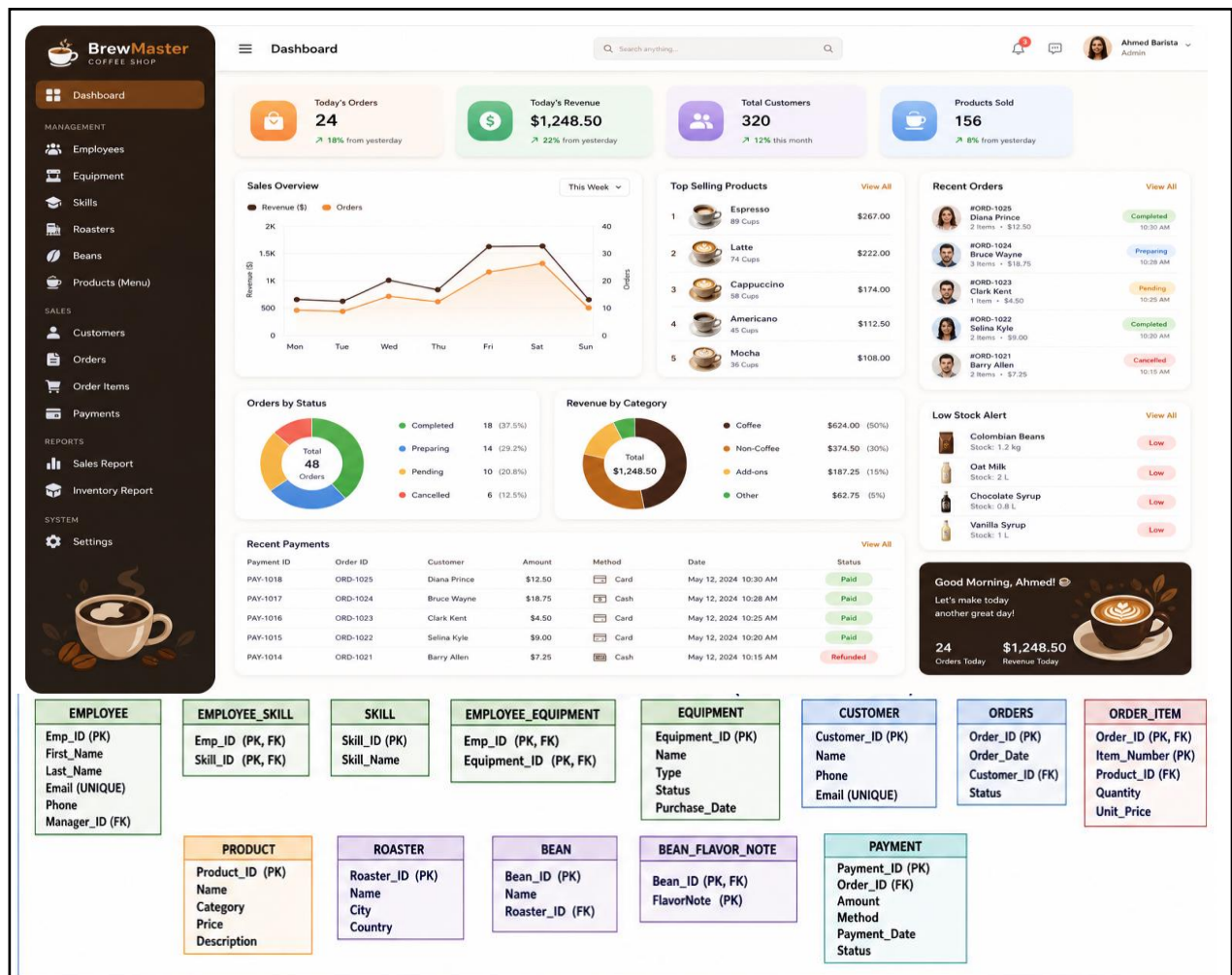


Figure 1: Database Structure Prototype

8. Constraints

• 8.1 some limitations of this project include:

- Limited time for development
- No real customer data
- Focus on database only (no full application)
- Basic tools used

• 8.2 Data Quality: Ensuring data consistency across tables

• 8.3 Resources: Limited hardware/software resources

• 8.4 Schedule: Completion within business timeline

9. Project Timeline

• 9.1 Project management time line include as shown below:

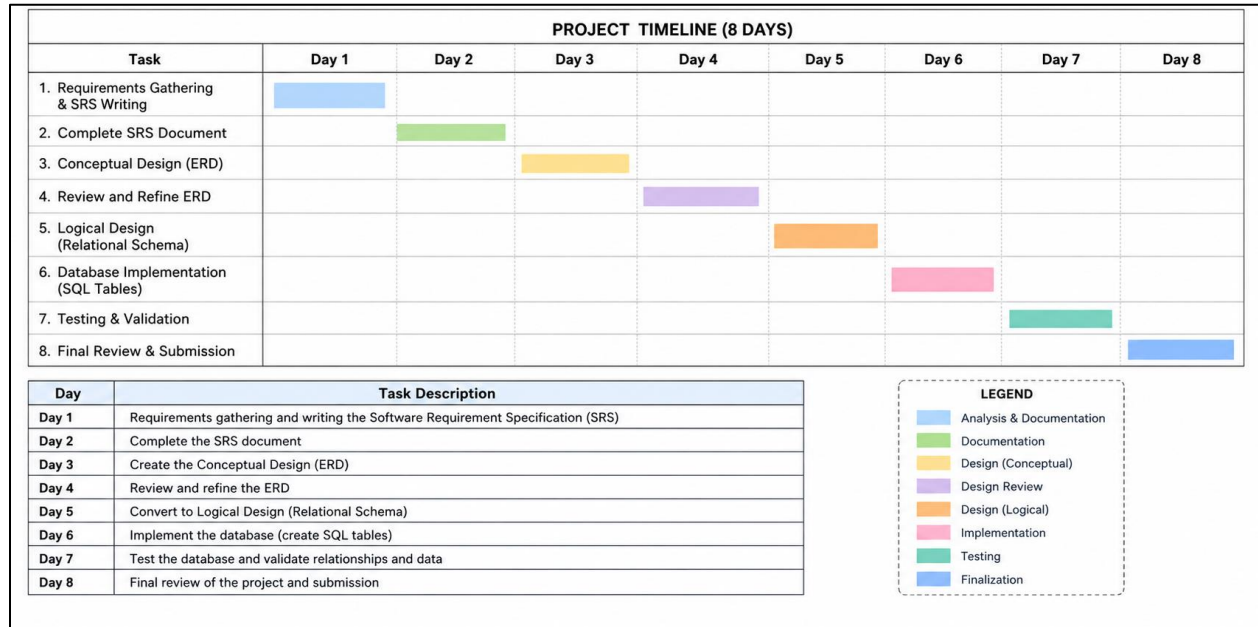


Figure.2 Project management time line

10. Financial

- The project requires resources related to database development and design tools.
- The main investment is the effort spent on designing the database and organizing data.
- The system provides value by improving data management and organization.
- Future development (such as dashboard or application) may require additional cost.

• 10.1 Costa Breakdown

Item	Description	Cost (OMR)
Database Design	Creating ERD and schema	100
Implementation	Building database tables	80
Testing & Validation	Checking relationships	50
Documentation	Writing report	30
Total		260 OMR

• 10.2 Payement Termes

- 50% upfront
- 50% after completion

11. Insights

• 11.1 at the end of the project, the analysis is expected to provide:

- Organized database structure
- Clear relationships between data
- Better understanding of coffee shop operations
- Reduced data redundancy
- Efficient data retrieval
- Support for future system expansion

12. Recommendations

• 12.1 Based on the expected analysis results, the project may recommend:

- Adding customer management system
- Developing a user interface (UI)
- Creating a dashboard for analysis
- Integrating payment system
- Expanding database features